Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/17/2023
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Probing the Effect of Simulated Microgravity on the Pathogenic Potential of Cariogenic Streptococcus mutans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/25/2022 per NSSC information (Ed., 4/27/22).		
Key Personnel Changes/Previous PI:	August 2021: No key personnel changes to report.		
COI Name (Institution):	Shields, Robert Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC18K1463		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Streptococcus mutans is a member of the human oral microbiota and a major etiological agent of dental caries (cavities) due to its ability to form biofilm, adapt to oral cavity stresses, and rapidly ferment carbohydrates. These characteristics promote its predomination of dental plaque biofilm and combined with repeated cycles of fermentative acid production, leads to tooth demineralization and cavity formation. Using High Aspect Rotating Vessels (HARVs) in the low-shear modelled microgravity (LSMMG) model, we previously demonstrated that S. mutans displays several physiological changes in response to simulated microgravity, including altered cell aggregation, enhanced sensitivity to oxidative stress, and global alterations in gene expression. Furthermore, simulated microgravity exposure has been shown by others to cause increased mandibular and alveolar bone loss and decreased saliva flow, factors that could predispose astronauts to caries. Therefore, the goal of this application is to characterize the physiological and genetic response of S. mutans to simulated microgravity, focusing specifically on biofilm development, stress resistance, and identifying key gene(s) and/or the core transcriptomic response that favors S. mutans survival during microgravity growth. This will be accomplished by three objectives: (1) S. mutans will be cultured in biofilm-promoting media under simulated microgravity conditions using the LSMMG and random-positioning machine (RPM) ground-based models of growth. Changes in cell aggregation, stress resistance, and gene expression profiling (via RNAseq) will be assessed with respect to normal gravity control cultures, and these results will be compared using bioinformatics and statistical techniques. This analysis will allow us to make more refined predictions as to which physiological and gene expression changes are likely to occur in S. mutans during spaceflight, and how this may impact its pathogenic potential; (2) Changes in S. mutans biofilm development in response to simulated microgravity will be studied using 2 ground-based models (LSMMG, RPM). S. mutans will be assessed in these models for biofilm viability and ultrastructure, exopolysaccharide production, and for its ability to compete with non-cariogenic oral streptococci; (3) A S. mutans Tn-seq library will be screened for mutants that lose competitive fitness during LSMMG and RPM-based simulated microgravity growth. This will be accomplished by using high-throughput DNA sequencing of pooled batches of Tn mutants pre- and post-HARV growth under both simulated microgravity and normal gravity conditions. This research is directed at understanding the physiological and genetic response of S. mutans to microgravity, especially with respect to multicellular behavior (biofilm formation), stress resistance, and the genes that favor S. mutans survival during microgravity growth. Therefore, this application directly addresses guiding questions MB-1 (What underlying genetic, molecular and biochemical processes are influenced by the space environment?), MB-3 (How does spaceflight influence microbial biofilm formation and microbial community development?), and MB-4 (What are the mechanisms that effect changes such as the altered virulence or altered drug resistance observed in some organisms during spaceflight?) of the 2016-2025 Space Biology Science Plan, each of which aligns with Priority P1 (multigenerational studies of microbial populations) of the National Research Council's (NRC) 2011 Decadal Survey Report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Collectively, the research proposed in this application is directed at understanding the physiological and genetic response of S. mutans to simulated microgravity, especially with respect to multicellular behavior (biofilm formation), stress resistance, and the underlying genetics that promote S. mutans fitness and growth. It is anticipated that the collective data generated from these three objectives will allow us to make better predictions as to which physiological and gene expression changes are likely to occur in S. mutans during spaceflight, and how this may impact its disease-causing potential and effect on astronaut health. The data generated in objectives 1-3 of this application will also contribute to our understanding of the molecular mechanisms by which S. mutans and other oral bacteria contribute to dental caries (cavities), since the low-shear force experienced during HARV simulated microgravity growth is very similar to that of saliva flow in the oral cavity.
Task Progress:	Overview/summary of research findings: • Certain elements of Streptococcus mutans physiology (growth properties, altered cell clumping) are conserved between the two ground-based models of simulated microgravity tested in this study (low-shear modelled microgravity (LSMMG) in the rotary cell culture system (RCCS), and the random positioning machine (RPM)) (Objective 1). • No significant differences in resistance to acid or oxidative stress were observed at mid-exponential or late-exponential growth phase in cultures grown in both the RCCS and RPM simulated microgravity models (Objective 1). • Relatively few significant ($P < 0.05$, no fold-change cutoff applied) transcriptome changes were observed by RNASeq between each simulated microgravity model (LSMMG, RPM) and the normal gravity RCCS control at mid-exponential (~3 hour) growth phase. This could be due to true inherent differences in these simulated microgravity models, or possibly because the RCCS normal gravity control is not an optimal control for the RPM model (Objective 1). • Many more significant ($P < 0.05$, 1.5 fold-change cutoff applied) transcriptome changes were observed by RNASeq in the RPM/normal gravity RCCS comparison, relative to the LSMMG/normal gravity RCCS comparison at late-exponential (~6 hour) growth phase. This suggests that simulated microgravity-induced gene expression changes in these models may not occur until later in the growth phase (Objective 1). • S. mutans and S. gordonii simulated microgravity (0 g) biofilms undergo significant structural changes relative to 0.9 g and 1 g control biofilms (Objective 2). • Confocal biofilm imaging using the RPM model is a useful and non-invasive technique to study the effects of simulated microgravity on bacterial biofilm formation (Objective 2). • Unexpected technical difficulties, combined with COVID-linked delays – shutdown of research activities at the University of Florida (UF) and NASA in 2020 – and subsequent timing of the Microgravity Simulation Support Facility (MSSF) renovations at NASA Kennedy Space Center (KSC), did not allow us to complete the proposed TnSeq experiments in Objective 3. Therefore, efforts were redirected at testing individual S. mutans mutants of interest in the RCCS model. These initial studies indicate that lrgAB may be of interest for follow-up studies, as growth of this mutant was reduced in the RCCS model, and this operon is known to be involved in pyruvate transport and oxidative stress resistance. We also conducted a third RNASeq study comparing gene expression in S. mutans and S. gordonii grown at simulated 0 g and 0.9 g on the RPM, based on the biofilm results observed in Objective 2.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Abstracts for Journals and Proceedings	Rice KC. "Direct imaging of dual-species Streptococcus mutans & Streptococcus gordonii biofilms grown under simulated microgravity." Gordon Research Conference on Streptococcal Biology, Newry, ME, August 14-19, 2022 Abstracts. Gordon Research Conference on Streptococcal Biology, Newry, ME, August 14-19, 2022. , Aug-2022

Abstracts for Journals and Proceedings	Davis KA, Hauserman MH, Rice KC. "Direct imaging of oral streptococcal & Staphylococcus aureus biofilms grown under simulated microgravity." Faculty Poster Session. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022 Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022 , Nov-2022
Abstracts for Journals and Proceedings	Davis KA, Rice KC. "Comparison of transcriptome changes of Streptococcus mutans cultured using two different ground-based simulated microgravity models." Graduate Poster Session. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts, 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Articles in Peer-reviewed Journals	Rice KC, Davis KAT. "Brief Communication: Confocal microscopy of oral streptococcal biofilms grown in simulated microgravity using a random positioning machine." npj Microgravity. 2024 Sep 9;10(1):1-5. https://doi.org/10.1038/s41526-024-00427-y , Sep-2024